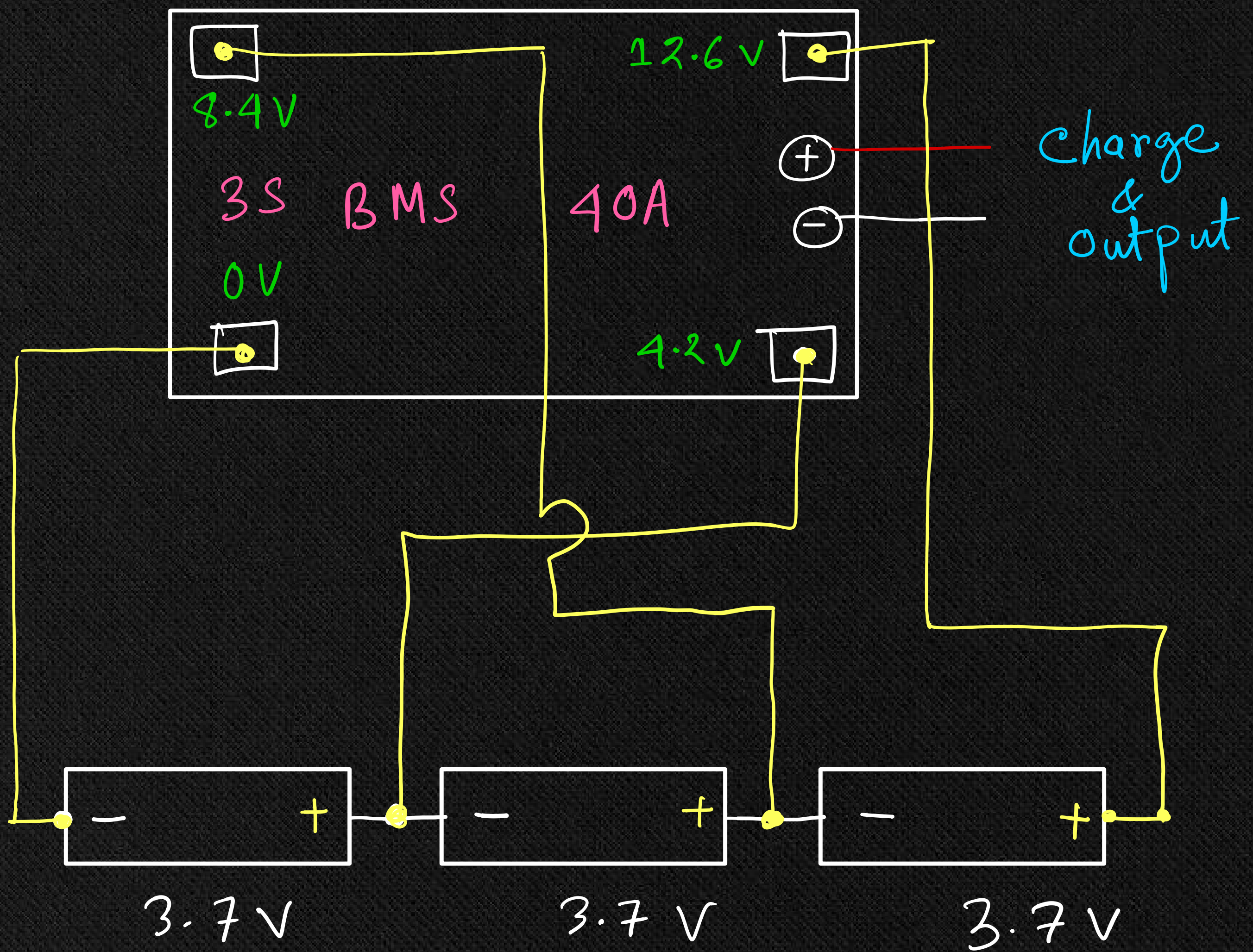
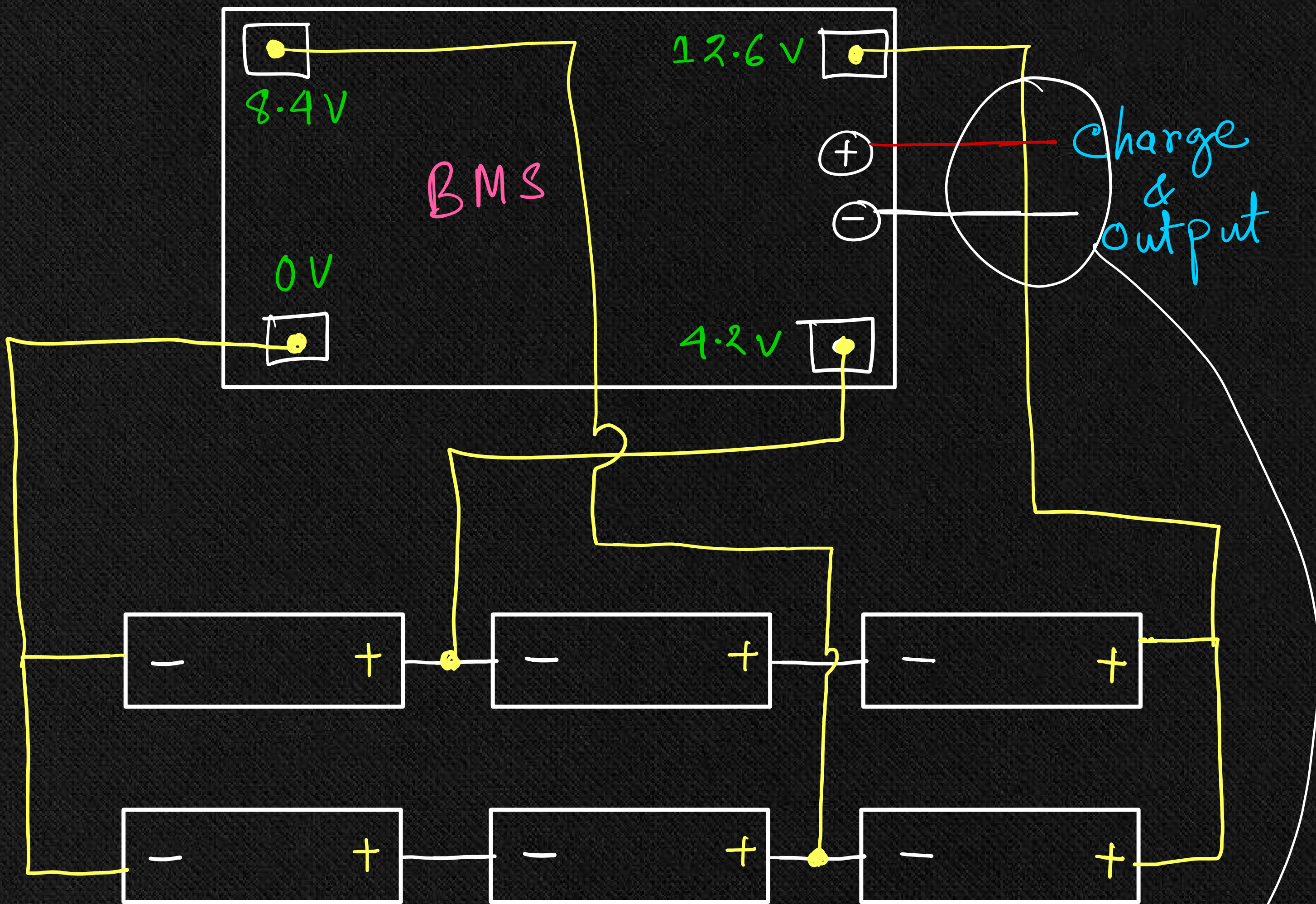


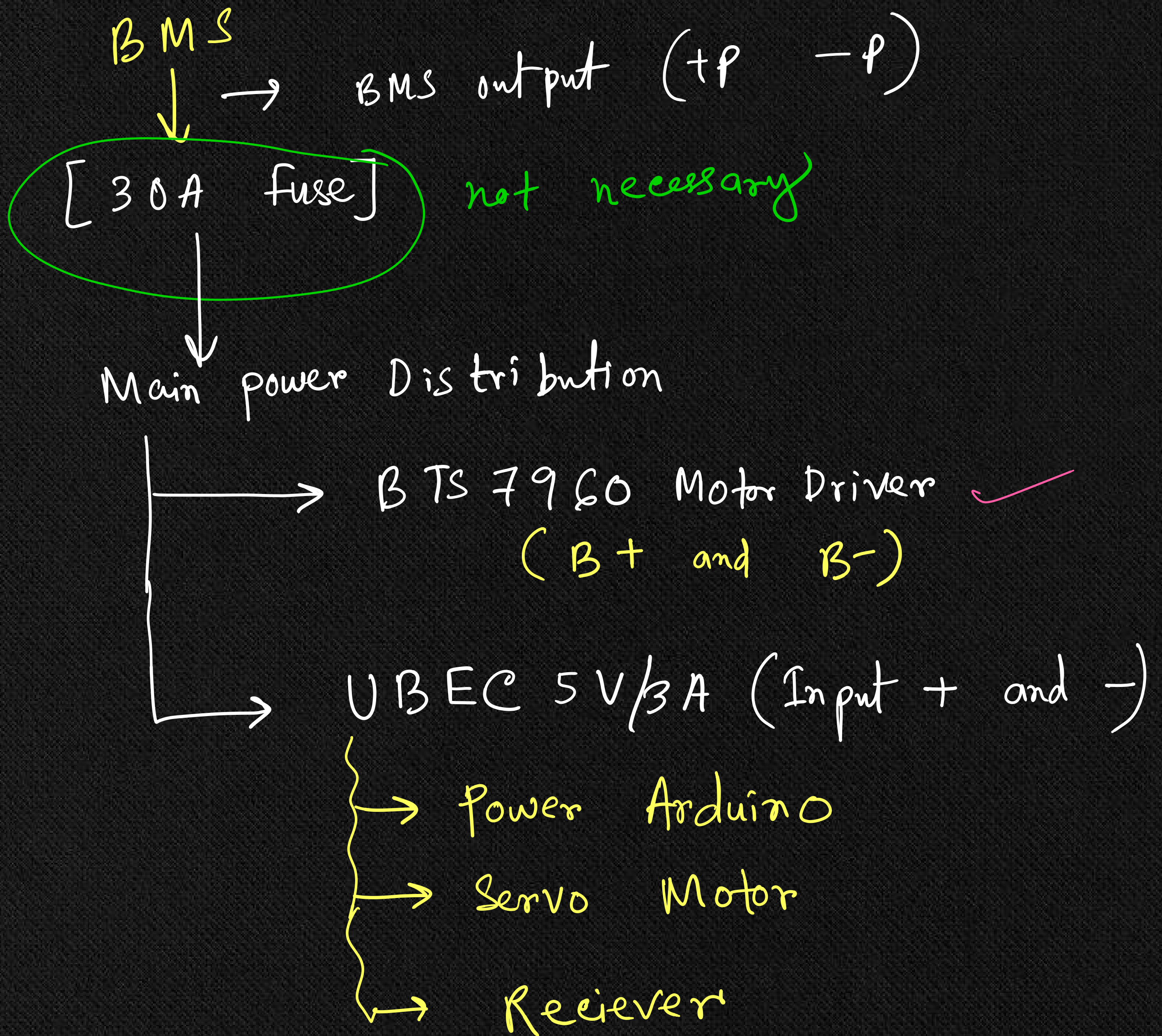
3S BMS Connection with Battery & XT60



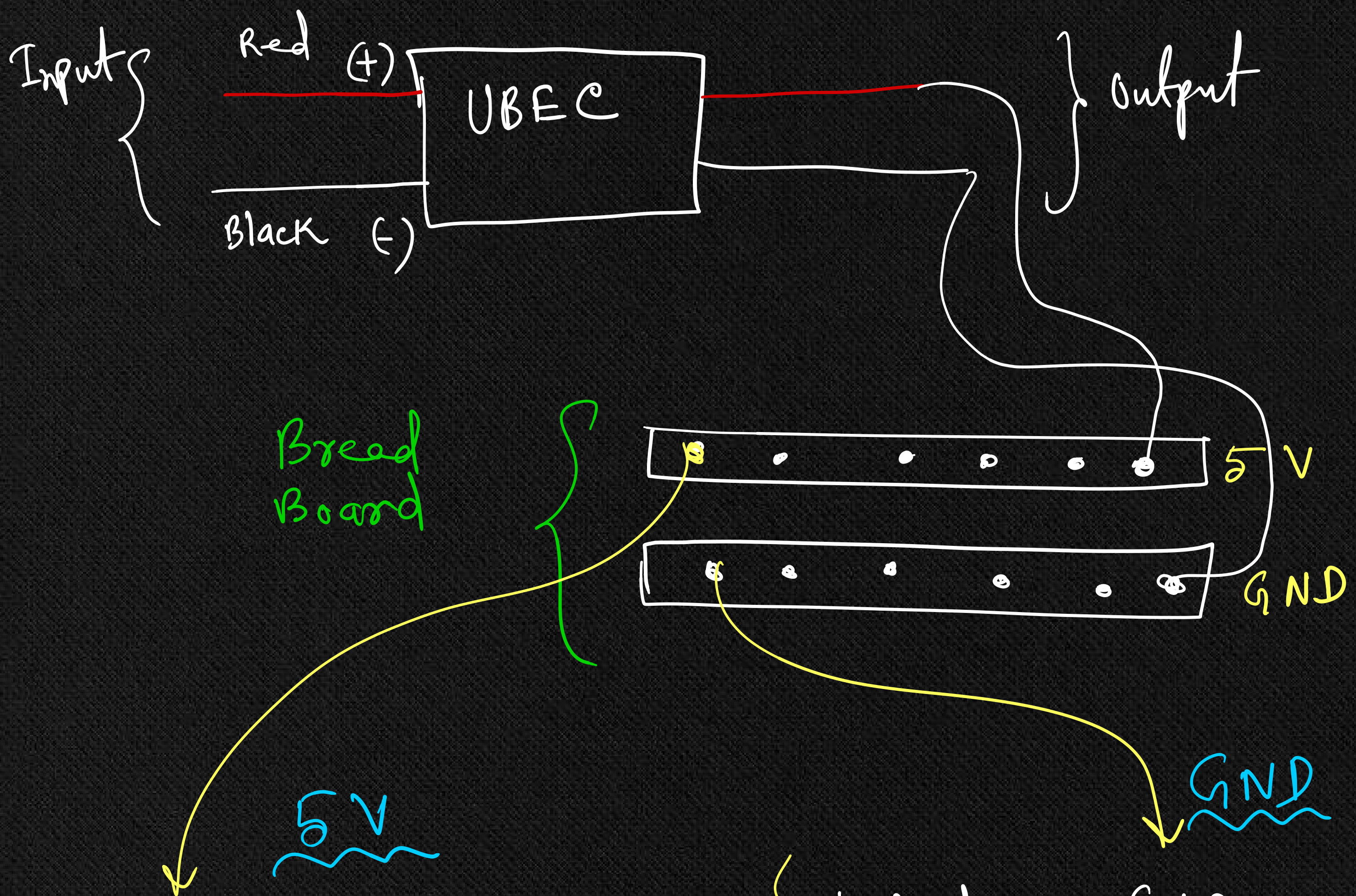
$$\Rightarrow 3.7 + 3.7 + 3.7 = 11.1 \text{ V}$$



Connection of BMS to Other Components



UBEC Circuit Connection



- * Arduino 5V pin
- * Servo Motor Red wire
- * Receiver (+) (middle)

- * Arduino GND pin
- * Servo Brown/Black wire
- * Receiver - pin (outer)

- All this connection are parallel - the UBEC's single 5V output feeds all three devices simultaneously
- Keep the wires as short as practical to avoid voltage drop.

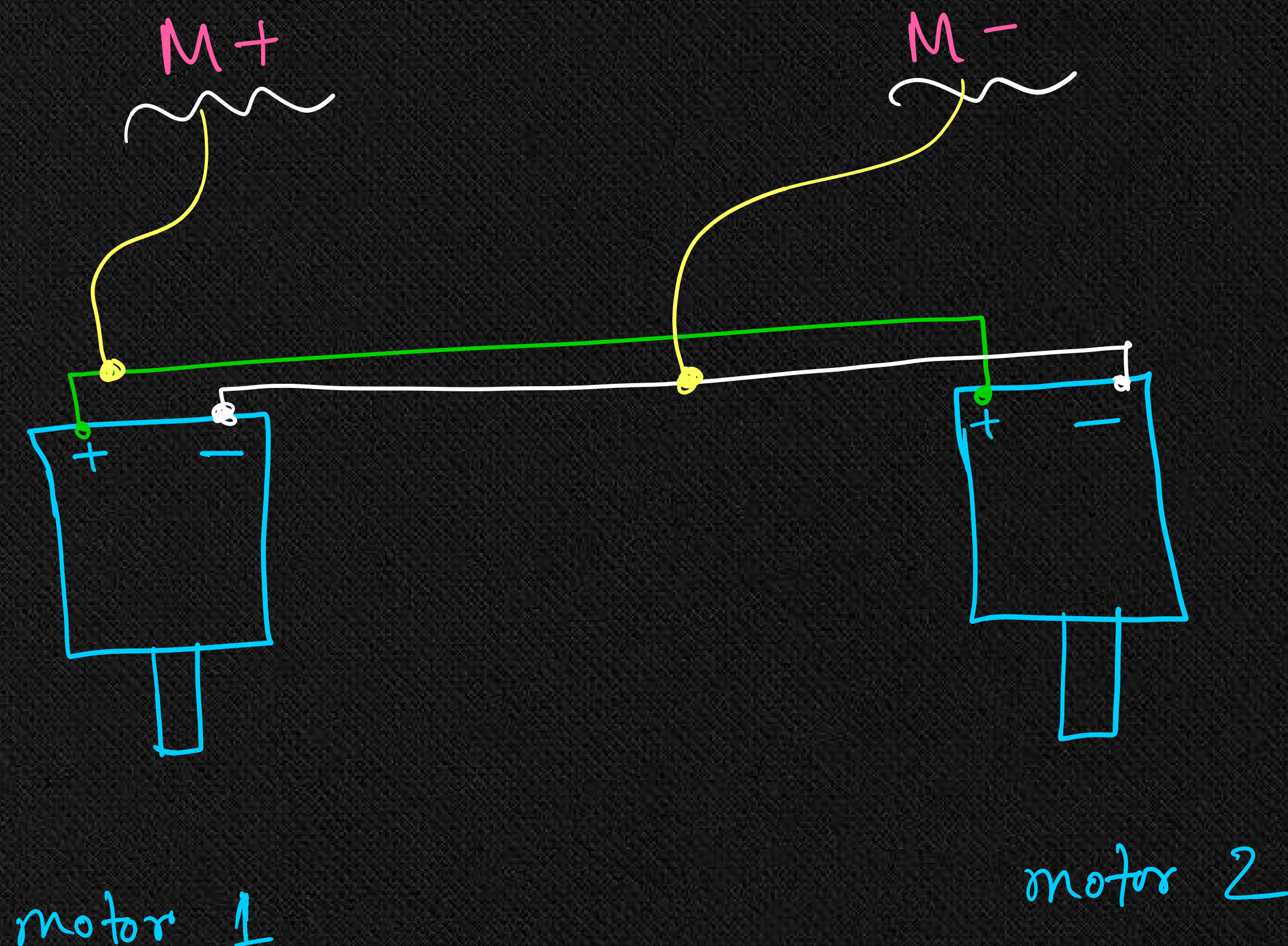
Arduino All pin Connections

Arduino Pin Connections (Summary Table)

Pin	Connected To	Wire Color	Purpose
2	FlySky Receiver CH1 (Throttle)	Yellow/White	Throttle signal input (Interrupt)
3	FlySky Receiver CH2 (Steering)	Yellow/White	Steering signal input (Interrupt)
5	BTS7960 RPWM	Any	PWM for forward motor control
6	BTS7960 LPWM	Any	PWM for reverse motor control
7	HC-SR04 Trig (optional)	Any	Ultrasonic trigger pulse
8	HC-SR04 Echo (optional)	Any	Ultrasonic echo receive
9	Servo Signal (orange/white)	Orange/White	Steering servo control
5V	UBEC 5V Output	Red	Powers Arduino (via 5V pin)
GND	UBEC GND	Black	Common ground (all devices)

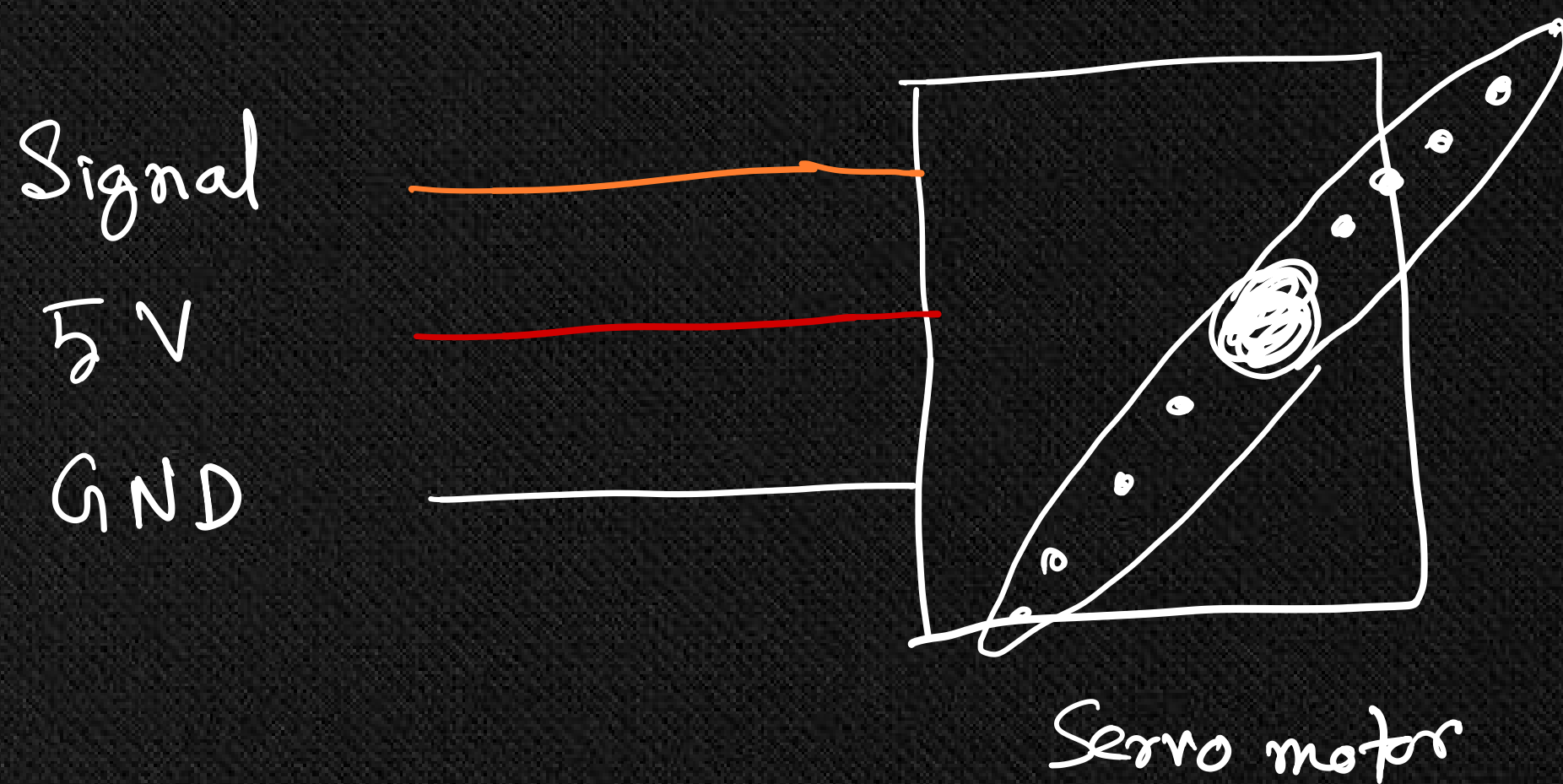
BTS 7960 Connection

BTS7960 Pin	Connect to	Notes
RPWM	Arduino Pin 5	PWM for forward speed/direction 1 10 .
LPWM	Arduino Pin 6	PWM for reverse speed/direction 1 10 .
R_EN	+5V (UBEC or Arduino 5V pin)	Must be HIGH to enable the driver 1 6 8 .
L_EN	+5V (UBEC or Arduino 5V pin)	Must be HIGH to enable the driver 1 6 8 .
VCC	+5V (UBEC or Arduino 5V pin)	Logic supply for the module's internal circuitry 6 10 .
GND	Arduino GND & Battery GND	Common ground for logic and power 4 6 10 .
B+	Battery Positive (via 30A fuse)	Main motor power supply 4 6 .
B-	Battery Negative	Main power ground 4 6 .
M+	Motor Positive	Connect to your DC motor's positive terminal 6 8 .
M-	Motor Negative	Connect to your DC motor's negative terminal 6 8 .



Connection of Servo Motor

Wire color	Pin Function	Connect To
Orange/Yellow	Signal	Arduino pin 9 (PWM)
Red	Power (5V)	UBEC 5V output
Brown/Black	Ground	UBEC GND (and common ground with Arduino)



Receiver Connection

FlySky FS-i4X Receiver Pinout

The receiver has multiple channels. You'll use **Channel 1 (Throttle)** and **Channel 2 (Steering)** . The other channels (3-6) are optional for future expansion.

Pin	Channel	Function	Connect To
1	CH1	Throttle (Up/Down)	Arduino Pin 2 (Interrupt 0)
2	CH2	Steering (Left/Right)	Arduino Pin 3 (Interrupt 1)
3	CH3	Auxiliary (optional)	Not connected (or future use)
4	CH4	Auxiliary (optional)	Not connected (or future use)
5	CH5	Auxiliary (optional)	Not connected (or future use)
6	CH6	Auxiliary (optional)	Not connected (or future use)
+	Power	5V Input	UBEC 5V Output
-	Ground	Ground	UBEC GND (common ground)

- Power: Connect UBEC 5V to the + pin and UBEC GND to the - pin of any one receiver port.
- Signal: Connect Arduino Pin 2 to CH1 (Signal), and Arduino Pin 3 to CH2 (Signal).
- Common Ground: Ensure the UBEC's GND is also connected to your Arduino's GND to complete the signal loop.

Steering

max = 1878

min = 1093

Thm

$$\max = 1813$$

$$\min = 1120$$